KK-G-486E
December 21, 1984
SUPERSEDING
Fed. Spec. KK-G-486D
July 20, 1976

FEDERAL SPECIFICATION

GLOVES, WELDER'S, LEATHER, GAUNTLET

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers horsehide and cattlehide work gloves used by welders, but does not include protective or safety gloves in which a rubber insert is used.

1.2 Classification.

1.2.1 Types and sizes. The gloves shall be of the following types and sizes as specified (see 6.2):

Type I - Welder's Gloves, Montpelier pattern (see figure 1).
Type II - Welder's Gloves, Gunn pattern (see figure 2).

Schedule of sizes

Extra, Extra-Small
Extra-Small
Small
Medium
Large
Extra-Large

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 8415

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Federal Specifications:

V-T-276	-	Thread, Cotton
KK-L-2004	-	Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
UU-P-268	-	Paper, Kraft, Wrapping
CCC-C-419	-	Cloth, Duck, Cotton, Unbleached, Plied -Yarns, (Army and Numbered)
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

Federal Standards:

FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-191	-	Textile Test Methods
FED-STD-311	-	Leather, Methods of Sampling and Testing
FED-STD-751	-	Stitches, Seams, and Stitchings

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from the specification may appear in the sample in which case the specification shall govern.

3.2 Material.

3.2.1 Leather. The leather for the components of the glove shall be as indicated in 3.7.1 and 3.8.1. When grain horsehide leather is used, the leather shall conform to type VII of KK-L-2004. When grain cattlehide leather is used, the leather shall conform to type V of KK-L-2004. When cattlehide splits are used, the leather shall conform to type VIII of KK-L-2004.

3.2.1.1 Thickness. The thickness of leather glove parts shall be not less than 2-1/2 nor more than 3-1/2 ounces. (Note: An ounce is equivalent in thickness to 1/64 inch.)

3.2.1.2 Color. The color of all leather components shall be cream or light grey. All leather components used in each pair of gloves shall be of the same color.

3.2.2 Thread cotton. Thread for sewing the gloves shall be unbleached cotton thread conforming to type IA1 or IB1 of V-T-276. When stitch type 301 is used, the thread shall conform to ticket No. 16, 4 ply. When stitch type 401 is used, the top thread shall be ticket No. 16, 4 ply and the bottom thread shall be ticket No. 36, 4 ply.

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3.2.3 Knitted wool-cotton cloth. The back lining shall be a knitted wool-cotton plaited cloth. The face yarn shall be woolen-spun yarn of 98.0 percent wool of not less than 50's grade (U.S. Standard) fleece or pulled wool. No reused or reprocessed wool or card waste or strippings shall be used. The backing yarn shall be a suitable cotton yarn and shall be not more than 30.0 percent of the weight of the cloth. The cloth shall be thoroughly scoured. The finished cloth shall conform to the following:

Weight, ounces per square yard	9 to 11
Wales	17 to 20
Courses, per inch	17 minimum
Color	Unbleached, light grey, or tan

3.2.4 Cotton duck. The cotton duck cloth shall conform to type III of CCC-C-419 and shall have a weight of not less than 8.25 ounces per square yard.

3.2.5 Label. Each glove shall have a combination size and identification label conforming to type III or IV, class 4 of DDD-L-20. The marking medium shall be yellow or black and shall show fair colorfastness to accelerated laundering when tested as specified in DDD-L-20. The type glove (type I or type II, as applicable) shall be included as part of the nomenclature when marking the glove.

3.3 Stitches, seams, and stitching. All stitching and seaming shall be as specified in table II and shall conform to FED-STD-751. When two or more stitch types are specified for the same operation either one may be used. The minimum and maximum number of stitches per inch shall be as specified in table II.

3.3.1 Type 301 stitching. Ends of 301 stitching shall be backstitched or overstitched 1/2 inch minimum except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so there will be no loose stitching resulting in a loose bobbin or top thread, or no excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn. Thread ends shall be trimmed to a length of not more than 1/2 inch.

3.3.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching.

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch back of the defective area, continue over the defective area and continue a minimum of 1/2 inch beyond the defective

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area onto the existing stitching. Loose or tight stitching shall be repaired by removing the defective stitching without damaging the material and restitching in the required manner. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

3.3.2 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching. Both ends of all seams or stitching produced with 401 stitch type, when not turned under in a hem or held down by other stitching shall have a 1/4 to 3/4 inch chain extending beyond each end. All repairs shall be in accordance with 3.3.1.1 a and 3.3.1.1 b. The looper thread shall be on the inside of the glove.

3.4 Dies and patterns. The dies or patterns for cutting all parts of the gloves shall be furnished by the contractor and shall be of the proper proportions to provide good fitting gloves conforming to finished measurements in table I.

3.5 Figures. Figures 1 and 2 are furnished for information purposes only. If there are any inconsistencies between the figures and the written specification, the written specification shall govern.

3.6 Dimensions of gloves. The finished gloves shall conform to the dimensions shown in table I. Measurements shall be taken as shown on figures 1 and 2.

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TABLE I. Finished dimensions of gloves

	SIZE						
	Extra, Extra small	Extra Small	Small	Medium	Large	Extra large	Tolerance plus or minus
	Inches	Inches	Inches	Inches	Inches	Inches	Inch
A - Top edge of cuff to tip of second finger	12-1/4	12-1/2	12-3/4	13	13-1/2	14	1/2
B - Bottom edge of cuff to tip of second finger	7-1/4	7-1/2	7-3/4	8	8-1/2	8-3/4	1/4
C - Across palm at thumb crotch, seam to seam	4	4-1/4	4-1/2	5	5-1/2	5-3/4	1/8
D - Tip of index finger to thumb crotch	4-1/4	4-1/2	4-3/4	5	5-1/2	5-3/4	1/8
E - Thumb crotch to tip of thumb	2-1/4	2-1/2	2-3/4	3	3-1/4	3-1/2	1/8
F - Across bottom edge of cuff	4-1/4	4-1/2	4-3/4	5	5-1/2	6	1/4
G - Across top edge of cuff	5-3/4	6	6-1/4	6-3/4	7	7-1/4	1/4

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3.7 Type I welder's gloves, Montpelier pattern.

3.7.1 Leather (components).

3.7.1.1 Front. The palm, thumb, fourchettes and gauntlet piece shall be split-cattlehide leather.

3.7.1.2 Back. The one-piece continuous back shall be grain-horsehide leather or grain cattlehide leather.

3.7.1.3 Reinforcing welts. The welts shall be of grain horsehide or grain cattlehide leather.

3.7.1.4 Dimensions of reinforcements. The fourchette strips shall be 3/4 ± 1/16 inch wide. The welt strips for reinforcing seams shall be 1/4 ± 1/16 inch wide.

3.7.2 Design. Type I gloves shall be of the pattern known commercially as Montpelier, as shown on figure 1. The gloves shall be made of a front section, a back section, and finger fourchettes. The front section shall be composed of three pieces, consisting of a separate gauntlet piece, a face of thumb-palm-finger piece, and a back of the thumb piece. The back section shall be made in one continuous piece consisting of the gauntlet, back, and four fingers. The gauntlets of both the front and the back shall be lined with the cotton duck specified in 3.2.4. In addition, the back section shall be lined from the bottom edge of the gauntlet duck lining to the ends of the fingers with the knitted cloth specified in 3.2.3. The palm, thumb, and fingers of the front section shall be unlined.

3.8 Type II welder's gloves, Gunn pattern.

3.8.1 Leather. Type II gloves shall be of split cattlehide leather including welts. The welts shall be 1/4 ± 1/16 inch wide.

3.8.2 Design. Type II gloves shall be of the pattern known commercially as Gunn, as shown on figure 2. The face of the thumb, the palm, and the first (index) and fourth (little) fingers shall be made of one piece of leather. The back shall be of one piece and shall include the backs of the fingers and gauntlet. The palm sides of the second and third fingers shall be of one piece. The back of the thumb shall be of one piece. The gauntlet front shall be of one piece. The gauntlet cuffs (front and back) shall be fully lined with cotton duck specified in 3.2.4. There shall be a lining of the knitted cloth specified in 3.2.3 in the back extending from the bottom of the gauntlet facing to the tips of the fingers.

3.9 Construction. The gloves shall be made by and with the use of all manufacturing operations listed in table II. The contractor is not required to follow the exact sequence of operations listed. During the spreading, cutting,

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and manufacturing process, components having material defects or damages that are classified as defects in 4.2.3 and 4.2.4 shall be removed from production and replaced with non-defective and properly matched components.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Type I glove.						
	a. Join back of thumb to palm and face of thumb.						
	Join back of thumb to palm and face of thumb using inseam construction 1/16 to 1/8 inch stitch margin. A leather welt strip shall be caught in the entire seam.	301 or 401	SSj-1	7-10	16-4	16-4	
			SSj-1	7-10	16-4	36-4	
	b. Line front gauntlet.						
	The front gauntlet shall be lined with the cotton duck specified in 3.2.4. To facilitate hemming, leave a 1/2 ± 1/16 inch margin between the top edge of the lining and the top edge of the unhemmed leather gauntlet.						
	c. Join front gauntlet to palm and thumb.						
	Join front gauntlet piece and cotton duck lining to palm and thumb using inseam construction 1/16 to 1/8 inch stitch margin, catching a leather welt strip in the seam. The ends of the welt shall be caught in the closing seam.	301 or 401	SSj-1	7-10	16-4	16-4	
			SSj-1	7-10	16-4	36-4	
	d. Line back of glove.						
	The continuous one piece back shall be lined throughout. The gauntlet section shall be lined with the cotton duck specified in 3.2.4. The top edge of the gauntlet lining shall be 1/2 ± 1/16 inches from	301 or 401	LSa-2	7-10	16-4	16-4	
			LSa-2	7-10	16-4	36-4	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Type I glove. (cont'd)						
	d. Line back of glove. (cont'd)						
	the top edge of the unhemmed leather gauntlet. The remaining area of the back of glove shall be lined with the knitted cloth specified in 3.2.3. The knitted cloth shall be lapped under and joined to the bottom edge of the gauntlet lining with a double row of stitches 1/8 to 3/16 inch gage. The knitted cloth shall extend over the backs of the fingers and be tacked at the fingertips. The cotton back of the lining shall be next to the leather and the wool fleece face exposed.						
	e. Hem gauntlet.						
	The top leather edge of the gauntlet shall be turned in to form a 1/2 ± 1/16 inch hem. The leather shall be lapped over the gauntlet lining and single stitched 1/16 to 1/8 inch from raw edge of leather.	301	EFa-1	7-10	16-4	16-4	
	f. Join fourchettes to back and front of glove.						
	Join fourchettes to back and front of glove using inseam construction 1/16 to 1/8 inch stitch gage.	301	SSa-1	7-10	16-4	16-4	
		or 401	SSa-1	7-10	16-4	36-4	
	g. Join front to back.						
	Join front of glove to back of glove with inseam construction 1/16 to 1/8 inch stitch gage from top of glove to top of little finger. A welt strip shall	301	SSj-1	7-10	16-4	16-4	
		or 401	SSj-1	7-10	16-4	36-4	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Type I glove. (cont'd)						
	g. Join front to back. (cont'd)						
	be caught in the seam from top of little finger to within 1/2 inch of the top of finished glove, but shall not be caught in the hem. The lining shall be caught in the inseam stitching.						
	Join front and back of glove on the thumb side using inseam construction 1/16 to 1/8 inch stitch gage from top of first finger to top of glove. A welt strip shall be caught in the seam from top of first finger to within 1/2 inch of the top of finished glove, but shall not be caught in the hem. The lining shall be caught in the inseam stitching.	301 or 401	SSj-1 SSj-1	7-10 7-10	16-4 16-4	16-4 36-4	
	h. Seams which have leather welt.						
	The distance between the stitch line and the edge of the welt on the outside of the glove shall be not greater than 1/8 inch.						
	i. Marking.						
	The label (see 3.2.5) shall be applied to the inside of the back of each glove in the center, approximately 1 inch down from finished top edge.						
*	j. Forming.						
	The gloves shall be dampened, layed-off, and formed on steam or electrically heated boards to proper shape and appearance.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Type I glove. (cont'd)						
	k. Pairing.						
	The gloves shall be matched, paired and tacked together with thread or non-corrosive soft metal staples. Tacking shall be done with not more than one soft metal staple or not more than two stitches using a cord thread that will permit the pair to be pulled apart without tearing the material.						
2.	Type II glove.						
	a. Join back of thumb to palm and face of thumb.						
	Use procedure specified for type I glove in operation 1a.						
	b. Line front gauntlet.						
	Use procedure specified for type I glove in operation 1b.						
	c. Join front gauntlet to palm and thumb.						
	Lap front gauntlet and cotton duck lining 1/2 ± 1/16 inch over palm and back of thumb and join together with a double row of stitching using a 1/8 to 3/16 inch stitch gage.	301	LSa-2	7-10	16-4	16-4	
		or					
		401	LSa-2	7-10	16-4	36-4	
	d. Line back of glove.						
	Use procedure specified for type I glove in operation 1d.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Type II glove. (cont'd)						
	e. Hem gauntlet.						
	Use procedure specified for type I glove in operation 1e.						
	f. Join base of second and third fingers to palm.						
	Lap palm over base of second and third fingers, place leather flap down flat on top of palm, align the edge of flap nearest fingertips with edge of palm, then join fingers, palm, and flap together with double row of stitches 1/16 to 1/8 inch stitch gage. The flap shall be of sufficient width so when it is turned back toward the fingertips and over the joining seam, the free portion shall be 1/4 to 1/2 inch in width. The ends of the flap shall be 3/8 ± 1/8 inch from the closing seam.	301 or 401	LSa-2 LSa-2	7-10 7-10	16-4 16-4	16-4 36-4	
	g. Join front and back of glove.						
	Join front and back of glove using inseam construction and 1/16 to 1/8 inch stitch gage from top of glove on thumb side all around to top of glove on the other side. A leather welt shall be caught in the closing seam starting 1/2 inch from the top of the glove on one side and continuing all around to within 1/2 inch of the top of the glove on the other side.	301 or 401	SSj-1 SSj-1	7-10 7-10	16-4 16-4	16-4 36-4	
	h. Seams which have leather welt.						
	Use procedure specified for type I glove in operation 1h.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Type II glove. (cont'd) i. Marking, forming, and pairing. Use procedure specified for type I glove in operations 1i, and 1j, and 1k.						

NATICK Form 903
1 Dec 7

EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

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3.10 Workmanship. The finished gloves shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, testing shall be performed on components as specified in 4.2.1.1.

4.2.1.1 Knitted wool-cotton cloth testing. The knitted wool-cotton cloth shall be tested for the characteristics indicated in table III. The test methods listed in table III shall be in accordance with FED-STD-191, wherever applicable. The lot size shall be expressed in units of 1 yard. The sample unit shall be 1/2 yard full width. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (no. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. Knitted wool-cotton cloth tests

Characteristic	Requirement paragraph	Test method
Identification of face yarn	3.2.3	1100 1/
Identification of backing yarn	3.2.3	1200 1/
Percent of cotton in cloth	3.2.3	2100 2/
Weight of cloth	3.2.3	5401
Wales per inch	3.2.3	5070
Courses per inch	3.2.3	5070

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One hundred minus percent of wool (dry desized basis) equals percent of cotton backing yarn in cloth.

4.2.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or services for which in-process inspection has indicated nonconformance.

4.2.3 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of one glove. The sample unit shall be one glove and the selection shall be by pairs. Defects for pairing shall be classified as a single defect. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE IV. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Pairing	Not properly mated, e.g., not right and left of same size.	X	
	Not tacked together.		X
	Wide variation in color in the pair.		X
Color	Not as specified.	X	
	Color not uniform or shaded parts clearly noticeable.		X
Design	Any characteristic not in accordance with the specified requirements (unless otherwise classified herein).	X	
Cleanliness	Spots or stains clearly noticeable.		X
Material (leather)	Any through hole (except pinhole), cut, tear, or rip.	X	
	Three or more pinholes in a glove.		X
	Not cattlehide or horsehide (where required).	X	
	Not full grain (where specified).	X	
	Not split leather (where specified).		X
	Loose, spongy, or flanky.	X	
	Not soft and pliable.	X	
	Open vein seriously affecting serviceability.	X	
	Open vein affecting serviceability but not seriously.		X
	Grain damage (where applicable): Penetrating deeply into leather	X	
	Not penetrating deeply into leather.		X
	Any open scratch or grub holes.	X	
	Scratch or grub or tick marks not penetrating deeply into leather fiber.		X
	Deep fat wrinkles.	X	
	Any flesh cut affecting service-ability.	X	
	Any scar, brittle or bony area, weak or thin spot, or abraded area.	X	
	Any healed grub mark or other healed damage seriously affect-ing serviceability.	X	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Material (leather) (cont'd)	Any healed grub mark or other healed damage affecting serviceability but not seriously.		X
	Flesh side of grain leather or either side of split leather not smooth or contains excessively loose flesh more than 2 square inches in total area.		X
Finish (leather)	Finish applied to grain surface.	X	
Knitted back lining or gauntlet lining	Not type specified.		X
	Any hole, cut, tear, rip, or drop stitch.	X	
	Multiple floats, loose end, or slubs.		X
	Any broken or missing pick, end, course, or wale.		X
	Any mend in cloth (except restitching of seams).		X
Construction and workmanship, general (applicable to all components unless otherwise indicated herein)	Any component or operation omitted.	X	
	Any component or assembly damaged.	X	
	Any component misplaced not serving purpose intended.		X
	Any component not securely affixed, e.g., gauntlet loosely attached to back or palm of glove.	X	
	Any workmanship damage, e.g., scissor or knife cut.	X	
	Any mend in leather, e.g., patch, (not applicable to restitched seam repair).	X	
	Needle chew.	X	
Seams and stitching	Any open seam:		
	- in single stitched seam	X	
	- in both rows of double stitched seam	X	
	- in one row of a double stitched seam		X
	- repaired but not repaired as specified.		X

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitching (cont'd)	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur.		
	Not specified seam type.	X	
	Not specified stitch type.	X	
	Any row of stitching omitted.	X	
	Loose stitch tension resulting in a loosely secured seam.	X	
	Tight stitch tension resulting in cutting of leather.	X	
	Ends of stitching not secured as specified.		X
	Gage of stitching irregular or not as specified seriously affecting serviceability.	X	
	Gage of stitching irregular or not as specified affecting serviceability but not seriously.		X
	Any part caught in unrelated row of stitching.	X	
	One or two stitches per inch less than specified.		X
	Three or more stitches per inch less than specified.	X	
	More than the specified maximum number of stitches per inch damaging leather.	X	
	More than the specified number of stitches per inch but not damaging the leather.		X
Assembly detail	Glove not neatly laid off.		X
	Finger distorted or twisted affecting comfort of wearer.	X	
	Thread ends not trimmed throughout glove.		X
	Not properly trimmed or finished, e.g., lump or bulk in finger inseam because leather was not properly trimmed.	X	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Assembly detail (cont'd)	Not constructed with specified number of pieces of leather (except welting).	X	
	Welting or reinforcing flap, if required, missing or not caught by stitching.	X	
	Any welting strip end that finishes more than 5/8 inch from top edge of cuff.		X
	Gauntlet not fully faced or lined, or glove not lined where specified.	X	
Marking	Missing, incomplete, incorrect, not legible, not specified size, not in proper location, or not accomplished in the specified manner.		X

4.2.4 End item dimensional examination. The gloves shall be examined for conformance to the thickness of leather requirements in 3.2.1.1 and to the dimensions specified in table I. Any thickness or dimension that is not within the established tolerance shall be classified as a defect. The lot size shall be expressed in units of one glove. The sample unit shall be one glove and the selection shall be by pairs. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0

NOTE: Each leather part of the glove shall be checked for thickness. i.e. thumb, body of glove, and fourchette strips. Any of these parts that are not within the required thickness range shall be classified as a defect. Thickness shall be determined in accordance with the procedure for "Fabricated leather articles" of Method 1011 of FED-STD-311. Where glove parts are lined, the lining should be parted from the leather and the thickness of the leather should be taken doubled. This dimension is then divided by two.

4.2.5 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

4.2.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence,or method of application.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.1.1 Level A packing. Sixty pairs of gloves of one type and size only, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-P-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The gloves shall be arranged flat, four pairs in length (alternated fingertip to cuff), one pair in width and fifteen pairs in depth within a shipping container. Every other pair of gloves in each stack shall also be alternated fingertip to cuff. Inside dimensions of each

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container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. When applicable, the contents shall be immobilized by filling the voids with fiberboard filler pads or crumpled kraft paper. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.5. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.1.2 Level B packing. Sixty pairs of gloves of one type and size only, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. The gloves shall be arranged flat, four pairs in length (alternated fingertip to cuff), one pair in width, and fifteen pairs in depth within a shipping container. Every other pair of gloves in each stack shall also be alternated fingertip to cuff. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. When applicable the contents shall be immobilized by filling the voids with fiberboard filler pads or crumpled kraft paper. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.5. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.1.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in PPP-B-636, except that the inspection shall be in accordance with 4.2.5.

5.1.3 Commercial packing. Gloves shall be packed in accordance with ASTM D 3951.

5.2 Palletization. When specified (see 6.2), gloves, packed as specified in 5.1, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern 3 shall be used (see Appendix D of MIL-STD-147). Interlocking of loads shall be effected by reversing the pattern of each course.

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5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with FED-STD-123 or ASTM D 3951, as applicable.

5.3.2 Military requirements. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.3 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper lable 5 by 4 inches with the words "MIXED NSN's" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity of pairs and NSN's contained therein.

6. NOTES

6.1 Intended use. The gloves are for use by welders and personnel engaged in similar occupations, as authorized. Type II welder's glove (Gunn pattern) shall be used by all of the Military Services.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and sizes required (see 1.2.1).
c. Selection of applicable level of packing (see 5.1).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
e. When palletization is required (see 5.2).
f. Marking required (see 5.3).

6.3 Samples. For access to samples address the procuring office issuing the invitation for bids.

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MILITARY INTERESTS:

Custodians

Army - GL
Navy - NU
Air Force - 99

Review Activities

Army - MD
Air Force - 82
DLA - CT

User Activities

Navy - MC, MS, YD, CG, SH
Air Force - 45

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
APM
DOE
FPI
TVA

PREPARING ACTIVITY:

Army - GL

Project No. 8415-0410

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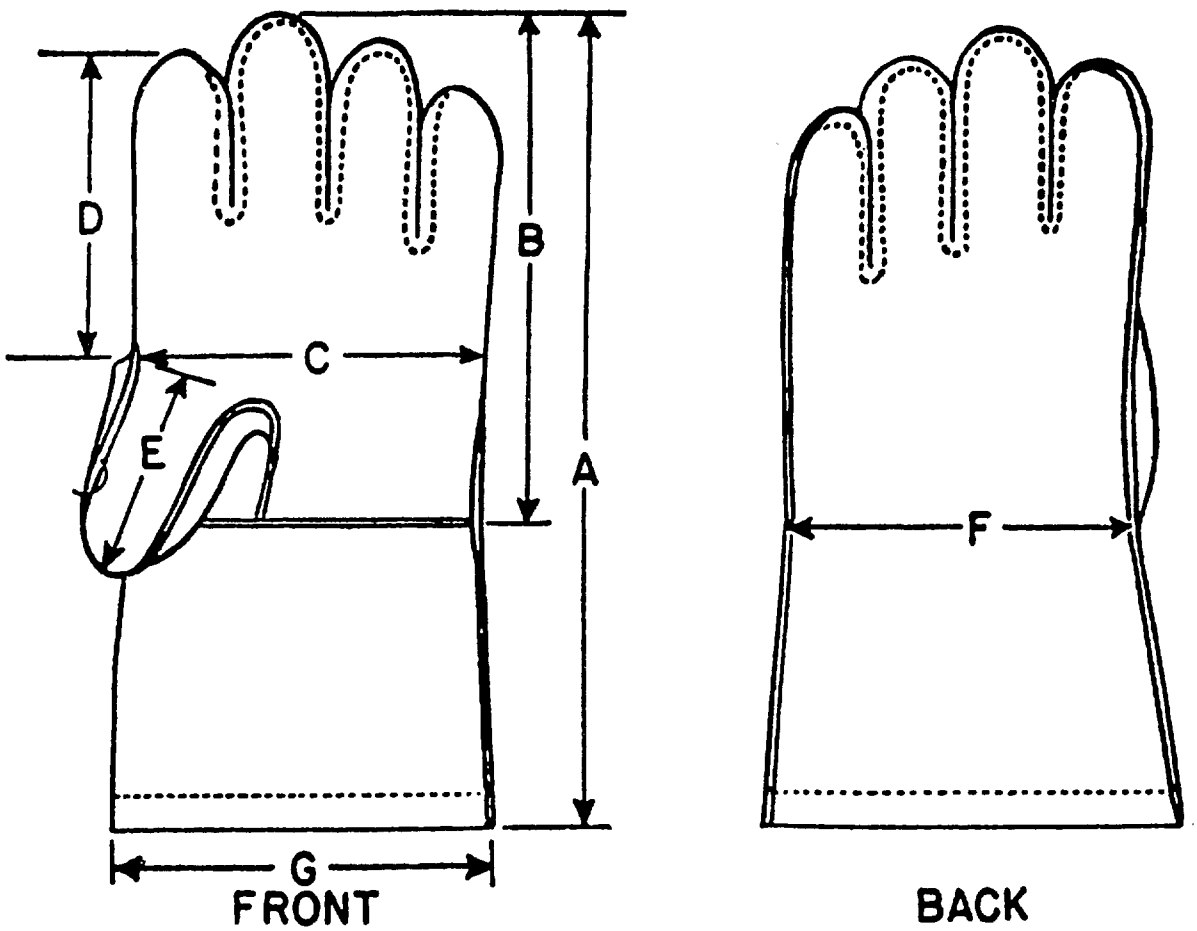


FIGURE 1. Type I, welder's gloves, Montpelier pattern.

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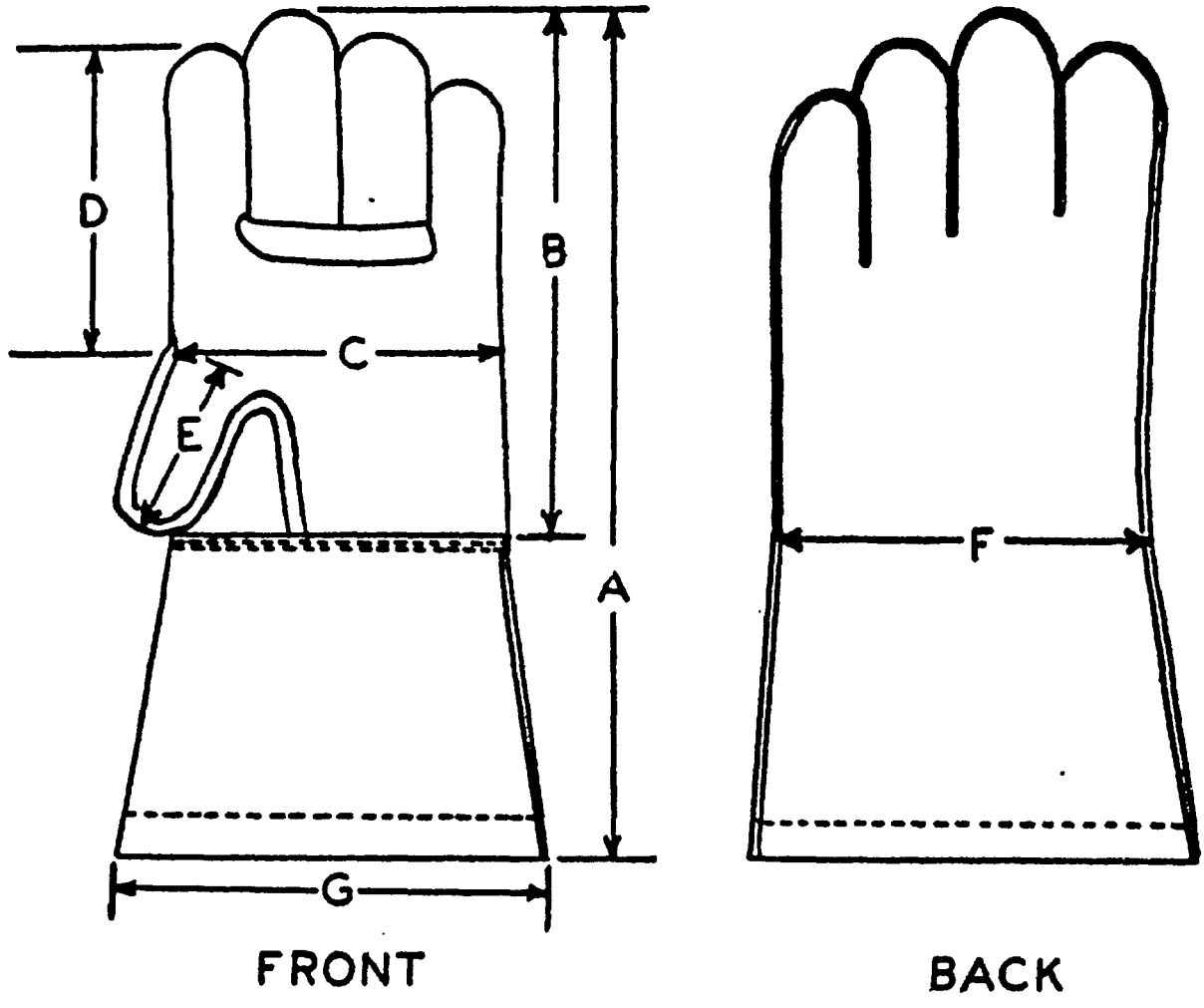


FIGURE 2. Type II, welder's gloves, Gunn pattern.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
KK-G-486E	Gloves, Welder's, Leather, Gauntlet

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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